Part 1

```python
# first 10 rows
print(df.head(10))

# df.info() df.describe()
print(df.info(), df.describe())
print("*" * 100)

# How many samples belong to each species?
print(df["species"].value_counts())
print("*" * 100)

# Are there any missing values?
print(df.isnull().sum())
```

[14] ✓ 0.0s

1.

```
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
118                7.7               2.6                6.9               2.3
122                7.7               2.8                6.7               2.0
117                7.7               3.8                6.7               2.2
105                7.6               3.0                6.6               2.1
131                7.9               3.8                6.4               2.0
107                7.3               2.9                6.3               1.8
130                7.4               2.8                6.1               1.9
109                7.2               3.6                6.1               2.5
135                7.7               3.0                6.1               2.3
100                6.3               3.3                6.0               2.5

     species  petal ratio  sepal area
118        2     3.000000       20.02
122        2     3.350000       21.56
117        2     3.045455       29.26
105        2     3.142857       22.80
131        2     3.200000       30.02
107        2     3.500000       21.17
130        2     3.210526       20.72
109        2     2.440000       25.92
135        2     2.652174       23.10
100        2     2.400000       20.79
```

2.

```
(150, 7)
<class 'pandas.core.frame.DataFrame'>
Index: 150 entries, 118 to 22
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    int64
 5   petal ratio        150 non-null    float64
 6   sepal area         150 non-null    float64
dtypes: float64(6), int64(1)
memory usage: 9.4 KB
```

```
memory usage: 9.4 KB
None        sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)     species  petal ratio  sepal area
count        150.000000  150.000000   150.000000  150.000000
mean           1.199333    1.000000     4.310500   17.822867
std            0.762238    0.819232     2.489648    3.361854
min            0.100000    0.000000     2.125000   10.000000
25%            0.300000    0.000000     2.802381   15.660000
50%            1.300000    1.000000     3.300000   17.660000
75%            1.800000    2.000000     4.666667   20.325000
max            2.500000    2.000000    15.000000   30.020000
```

3.

```
species
2    50
1    50
0    50
Name: count, dtype: int64
```

4.

```
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
petal ratio          0
sepal area           0
dtype: int64
```

part 2

```python
# Create New Columns:
# petal ratio = petal length / petal width
# sepal area = sepal length x sepal width
 (variable) df: DataFrame cal length (cm)"] / df["petal width (cm)"]
df["sepal area"] = df["sepal length (cm)"] * df["sepal width (cm)"]

# Filtering:All samples with sepal length > 6 / All samples from species 0 with petal width < 0.5
print(f"All samples with sepal length > 6:\n{df[df["petal length (cm)"] > 6]}\n")
print("*" * 100)
print(f"All samples from species 0 with petal width < 0.5:\n {df.loc[(df["species"] == 0) & (df["petal width (cm)"] < 0.5)]}")

# Sorting: Sort dataset by petal length (descending)
df = df.sort_values(by = "petal length (cm)", ascending = False)
```

0.0s Python

1.

```
All samples with sepal length > 6:
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
105                7.6               3.0                6.6               2.1
107                7.3               2.9                6.3               1.8
109                7.2               3.6                6.1               2.5
117                7.7               3.8                6.7               2.2
118                7.7               2.6                6.9               2.3
122                7.7               2.8                6.7               2.0
130                7.4               2.8                6.1               1.9
131                7.9               3.8                6.4               2.0
135                7.7               3.0                6.1               2.3

     species  petal ratio  sepal area
105        2     3.142857       22.80
107        2     3.500000       21.17
109        2     2.440000       25.92
117        2     3.045455       29.26
118        2     3.000000       20.02
122        2     3.350000       21.56
130        2     3.210526       20.72
131        2     3.200000       30.02
135        2     2.652174       23.10
```

```
**********************************************************************************************
All samples from species 0 with petal width < 0.5:
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                  5.1               3.5                1.4               0.2
1                  4.9               3.0                1.4               0.2
2                  4.7               3.2                1.3               0.2
3                  4.6               3.1                1.5               0.2
4                  5.0               3.6                1.4               0.2
5                  5.4               3.9                1.7               0.4
6                  4.6               3.4                1.4               0.3
7                  5.0               3.4                1.5               0.2
8                  4.4               2.9                1.4               0.2
9                  4.9               3.1                1.5               0.1
10                 5.4               3.7                1.5               0.2
11                 4.8               3.4                1.6               0.2
12                 4.8               3.0                1.4               0.1
13                 4.3               3.0                1.1               0.1
14                 5.8               4.0                1.2               0.2
15                 5.7               4.4                1.5               0.4
16                 5.4               3.9                1.3               0.4
17                 5.1               3.5                1.4               0.3
18                 5.7               3.8                1.7               0.3
19                 5.1               3.8                1.5               0.3
20                 5.4               3.4                1.7               0.2
21                 5.1               3.7                1.5               0.4
22                 4.6               3.6                1.0               0.2
24                 4.8               3.4                1.9               0.2
25                 5.0               3.0                1.6               0.2
26                 5.0               3.4                1.6               0.4
27                 5.2               3.5                1.5               0.2
```

```
   species  petal ratio  sepal area
         0     7.000000       17.85
         0     7.000000       14.70
         0     6.500000       15.04
         0     7.500000       14.26
         0     7.000000       18.00
         0     4.250000       21.06
         0     4.666667       15.64
         0     7.500000       17.00
         0     7.000000       12.76
         0    15.000000       15.19
0        0     7.500000       19.98
1        0     8.000000       16.32
2        0    14.000000       14.40
3        0    11.000000       12.90
4        0     6.000000       23.20
5        0     3.750000       25.08
6        0     3.250000       21.06
7        0     4.666667       17.85
8        0     5.666667       21.66
9        0     5.000000       19.38
```

part 3

```python
# Compute the mean of each feature grouped by species.
print(f"the mean of each feature grouped by species:\n{df.groupby("species").mean()}")
print("*" * 100)

# Compute both mean and standard deviation.
print(f"mean:/n{df.mean()}")
print(f"std:/n{df.std()}")
print("*" * 100)

# Which species has:Largest average petal length? Smallest average sepal width?
print(f"The species with the largest average petal length: {df.groupby("species").mean()["petal length (cm)"].idxmax()}")
print(f"The species with the smallest average sepal width: {df.groupby("species").mean()["sepal width (cm)"].idxmin()}")
print("*" * 100)

# Use pivot table to summarize mean petal length by species.
pivot_mean = pd.pivot_table(df, index='species', values='petal length (cm)', aggfunc='mean')
pivot_mean.rename(columns={'petal length (cm)': 'mean_petal_length (cm)'}, inplace=True)
print(pivot_mean)
```

[4] ✓ 0.0s 在 Data Wrangler 中打开"pivot_mean"

1.

```
the mean of each feature grouped by species:
         sepal length (cm)  sepal width (cm)  petal length (cm)  \
species
0                    5.006             3.428              1.462
1                    5.936             2.770              4.260
2                    6.588             2.974              5.552

         petal width (cm)  petal ratio  sepal area
species
0                   0.246     6.908000     17.2578
1                   1.326     3.242837     16.5262
2                   2.026     2.780662     19.6846
****************************************************************************
```

2.

```
****************************************************************
mean:/nsepal length (cm)     5.843333
sepal width (cm)      3.057333
petal length (cm)     3.758000
petal width (cm)      1.199333
species               1.000000
petal ratio           4.310500
sepal area           17.822867
dtype: float64
std:/nsepal length (cm)    0.828066
sepal width (cm)     0.435866
petal length (cm)    1.765298
petal width (cm)     0.762238
species              0.819232
petal ratio          2.489648
sepal area           3.361854
dtype: float64
```

3.

```
****************************************************************************
The species with the largest average petal length: 2
The species with the smallest average sepal width: 1
****************************************************************************
```

4.

```
*****************************************
         mean_petal_length (cm)
species
0                         1.462
1                         4.260
2                         5.552
```

part 4

```python
# Convert feature matrix to a NumPy array.
arr = df.to_numpy()

# Compute:Column-wise mean Column-wise standard deviation
print(f"Column-wise mean: f{np.mean(arr, axis = 0)}")
print(f"Column-wise std: f{np.std(arr, axis = 0)}")

# Reshape the array into shape (150, 4).
print(arr.shape)
# 前四列?
arr = arr[:, :4]
```

✓ 0.0s

1.

```
Column-wise mean: f[ 5.84333333  3.05733333  3.758       1.19933333  1.          4.31049976
 17.82286667]
Column-wise std: f[0.82530129 0.43441097 1.75940407 0.75969263 0.81649658 2.4813355
 3.35062866]
(150, 7)
```